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Prevention and Management of Pressure Injuries and Chronic Ulcers

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SUMMARY PEARLS

- Pressure injuries usually occur over a bony prominence but may also be related to a medical device or other object.
- Chronic ulcers are a manifestation of underlying disease processes.
- Peripheral arterial disease is common in individuals with diabetes and arterial insufficiency.
- Ischemic ulcers are typically below the ankle with a pale or gray, nongranulating base.
- Venous ulcers are typically located at the gaiter area of the calf or perimalleolar area of the lower extremity.
- Neuropathic ulcers are typically painless and rimmed by callus.
- Offloading, or the relief of a mechanical load, from the foot is recommended as part of the standard treatment of diabetic foot ulcers. Even with optimal local care, debridement, and dressings, a diabetic foot ulcer is unlikely to heal without consistent offloading.
- Reducing shear can be accomplished by positioning, such as raising the knees before raising the head of the bed to reduce the shear to the coccyx and sacral area, and by using antishear textiles, such as linens and underpads.

Introduction

Ulcer healing is a fine balance between biologic and molecular processes that involves hemostasis, inflammation, proliferation, and remodeling. To achieve ulcer healing, several factors need to be in place. A well-vascularized ulcer bed with an intact immune system is essential. Abnormal ulcer healing occurs when optimal conditions are absent, causing ulcers to move from an acute phase to a chronic phase.

Abnormal Ulcer Healing

Chronic ulcers, those that remain unhealed after 4 weeks, are generally stalled in the inflammatory phase and have a high incidence of biofilm-associated inflammatory mediators, such as interleukin-6 (IL6), IL10, and tumor necrosis factor-alpha (TNF-alpha), that can contribute to perpetual nonhealing and wound bed cellular senescence.[32] It is important to have an understanding of available treatment options to reactivate the healing cascade. In addition, nonhealing ulcers are a manifestation of underlying disease processes. Nonhealing ulcers most commonly result from (1) ischemia (arterial and small vessel disease), (2) venous insufficiency and associated lymphatic dysfunction (phlebolymphedema),[57] (3) neuropathy, and (4) pressure injury (Fig. 25.1).

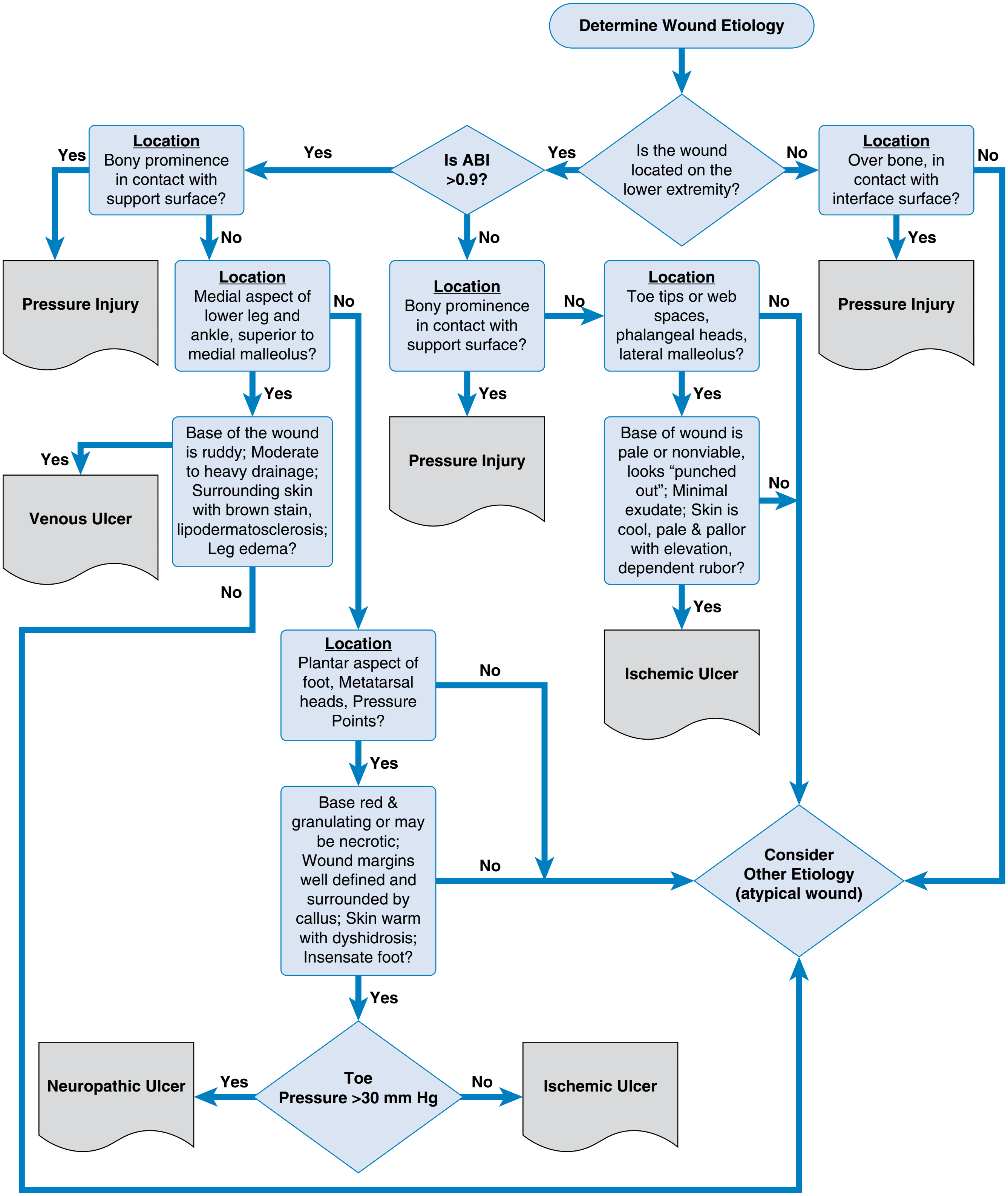
Ischemic Ulcers

Arterial

Ischemic ulcers typically result from a minor laceration, abrasion, or pressure in a limb with compromised blood flow. A shoe that is too narrow or has a poor-fitting toe box creates pressure-induced ischemia. This leads to a chronic, painful ulcer that forms from insufficient blood supply resulting in inadequate tissue oxygen and nutrients to meet the increased demands of healing. Ischemic ulcers typically occur in the distal extremity.[45] Ischemic and neuroischemic ulcers are common on the tips of the toes or the lateral borders of the foot.[29] Ischemic ulcers appear "punched out" with a sharply demarcated border (Fig. 25.2). The physical examination may also reveal decreased or absent peripheral pulses, lack of hair over the distal leg, cyanosis, skin atrophy, elevation pallor, and dependent rubor. When in doubt about the palpation of pedal pulses, obtaining an ankle brachial index (and ideally a toe brachial index in diabetics because of possible vascular calcification) is considered standard of care.[161] Ulcers below the ankle with a pale or gray, nongranulating base are usually the result of major arterial insufficiency. Simultaneous appearance of significant pain, livedo reticularis, and cutaneous toe infarcts (with subsequent ulceration) suggests atheromatous embolization. These conditions can also occur in periarteritis nodosa, systemic lupus erythematosus, or livedoid vasculitis.

Small Vessel

Small vessel ischemic ulcers are usually located on the leg and have areas of cutaneous infarction, which coalesce and proceed to ulceration.[92] Diagnosis of small vessel vasculitis and vasculopathies prompts the question as to the underlying cause. Although most think immediately of connective tissue disease, it is more commonly secondary to medications or infectious disease. Less



• **Fig. 25.1** Ulcer etiology algorithm. *ABI,* Ankle-brachial index.

• **Fig. 25.2** Arterial ulcer.

common causes of small vessel vasculitis include malignancy and Henoch-Schönlein purpura.[99] Rheumatology consultations can be very beneficial in establishing causation and therapeutic management.

Management

Ischemic ulcer therapy includes management of risk factors (all forms of tobacco cessation [including vaping], diabetes control, hypertension, hyperlipidemia) and restoration of peripheral blood flow (endovascular or surgical revascularization).[75] If arterial revascularization is not an option, arterial pumps may be of benefit for angiogenesis and improved wound healing.[120] In addition, management of ischemic ulcers involves conservative debridement, biofilm management,[120,168] pain control, and ulcer care. See Chapter 26 for details of the evaluation and management of ischemic (arterial and small vessel) disease.

Venous Ulcers and Associated Lymphatic Dysfunction Management (Phlebolymphedema)

A 2015 systematic review showed that in the United States, $3.5 billion are spent on lower extremity venous ulcers annually.[44] In patients with a history of chronic venous insufficiency, clinical findings include edema, hyperpigmentation, and soft tissue fibrosis.[47] Venous ulcers are typically located on the gaiter area of the calf or perimalleolar area of the lower extremity (Fig. 25.3). Upper extremity ulcerations are not caused by chronic venous insufficiency unless an arteriovenous fistula is present (typically located at the fingertips, especially if the fistula has been placed for dialysis).[56,92]

Management

Management guidelines have identified edema control as the cornerstone in the treatment of venous leg ulcers.[39,57,131,133] This can be achieved through leg elevation, compression therapy with wraps, stockings, or intermittent pneumatic compression.[136] In view of the high rate of recurrence, prescriptive fitted calf-high compression stockings therapy is the current standard of practice for the prevention of recurrence.[131] Oral pentoxifylline, as a rheologic agent, has been a standard of care for venous leg ulcers in the United States,[93,136] in addition to horse chestnut seed extract and other over-the-counter venotonics and lymphotonics.[136] Micronized purified flavonoid fraction (MPFF), derived from the peels of citrus fruits, has been a standard of care in Europe and Central America for decades, with greater than 50 peer-reviewed manuscripts and 12 randomized controlled trials (RCTs) with a 2008 meta-analysis supporting use for patients with venous disease.[94] MPFF was introduced into the United States in the past 10 to 15 years and is currently available over the counter. The active medicinal components of MPFF are diosmin and hesperidin among other lesser concentrations of flavonoids. Benchtop to clinical data demonstrates antiinflammatory properties (decreased intercellular adhesion molecule, vascular cell adhesion molecule, and TNF-alpha) improved venous and lymphatic tone as well as endothelial cell function with daily MPFF use.[112] MPFF use for venous disease was also recognized in the 2023 societal guidelines for venous interventionalists.[70] Advanced pneumatic compression devices, which provide automated manual lymphatic drainage, have been shown to be of benefit for phlebolymphedema when used within the context of prescriptive lower extremity compression and other components of complete decongestive therapy.[107,112] See Chapter 26 for details on the evaluation and management of venous disease.

• **Fig. 25.3** Venous ulcer.

Neuropathic Ulcers

Neuropathic ulcers form as a result of peripheral neuropathy, typically in diabetic patients or caused by an idiopathic cause. Loss of sensation, foot deformities, and limited joint mobility can result in abnormal biomechanical loading of the foot.[7,29,31,65,137] Lack of sensation over pressure points leads to microtrauma and eventual ulceration. Ulcers are typically located at the metatarsal heads; pulp or tip of a rigid great toe; or lateral surface of the foot at the base of the metatarsal, heel, or midfoot (in the area of a collapsed navicular or cuboid). Neuropathic ulcers are characteristically painless and rimmed by callus; they can be present for years before the patient seeks medical consultation (Fig. 25.4).

The frequency and severity of foot problems vary from region to region. Foot ulcers are a prevalent problem for people with

• **Fig. 25.4** Neuropathic ulcer.

diabetes, with a yearly incidence of around 2% to 4% in developed countries[21] and likely even higher in developing countries.[29] Data from the International Diabetes Federation shows 537 million adults worldwide have a diagnosis of diabetes mellitus in 2023, and this is predicted to rise to 700 million by 2045.[87] The frequency and severity of foot problems vary from region to region largely caused by differences in socioeconomic conditions, type of footwear, and standards of foot care.

Monteiro-Soares et al.[118] identified several observational studies where abnormal foot shape, rigid toe deformity, reduced first metatarsophalangeal mobility, and limited subtalar joint mobility were significantly associated with the development of diabetic foot ulcers. Common deformities in patients with diabetes include hammertoes, claw toes, prominent metatarsal heads, and hallux valgus.[66,150] Other structural abnormalities that may play a role in the development of diabetic foot ulcers include ankle equinus with restricted dorsiflexion,[102] the distal migration of the adipose tissue under the metatarsal heads,[73] and thickening of skin, tendons, ligaments, and joint capsules at the ankle and foot.[66] Foot deformities and limited joint mobility of the ankle and foot make the foot susceptible to abnormal areas of high pressure, restrict the foot's ability to absorb shock, and increase the risk of ulceration.[121] Charcot changes can be present, including medial tarsal subluxation, pronation, forefoot valgus, increased width, and decreased length. In patients with Charcot arthropathy or prior partial foot amputations, the biomechanics of the foot are more severely affected.[66] Neuropathy is a common complication of diabetes. This is characterized by a progressive loss of peripheral nerve fibers caused by decreased blood flow and high glycemic levels.[13,51] Endoneural hypoxia results in nerve fiber loss.[109,166] The duration and intensity of hyperglycemia strongly influence the severity of the neuropathy.[51]

Motor nerve damage is another mechanism through which distal symmetric sensorimotor polyneuropathy is associated with foot injuries. Motor neuropathy causes progressive atrophy of the intrinsic muscles of the foot leading to deformities. Highlighting the effect of motor neuropathy, one study showed that foot muscle atrophy may occur relatively early in the course of the disease before other clinical manifestations of neuropathy.[74] Autonomic neuropathy can also play a role in foot ulceration by causing hypohidrosis and dry skin that can easily crack and fissure.[65] Furthermore, autonomic neuropathy has been associated with functional abnormalities of the microcirculation and thermoregulatory capacity that may impair tissue perfusion and increase the risk of ulceration.[65,126] Once loss of protective sensation develops as a consequence of sensory neuropathy, patients become vulnerable to mechanical or thermal injuries.[150] In persons with neuropathy, minor trauma (ill-fitting shoes, walking barefoot, or an acute injury) can cause an ulcer. Ill-fitting shoes are a prevalent cause of diabetic foot ulcers and can potentially lead to amputation.[144] Neuropathic/traumatic ulcers result from normal walking pressures (40–60 pounds per square inch) for thousands of repetitions per day. In the 1950s, Naylor[129,130] demonstrated that a peak perpendicular load by itself is not necessarily harmful. He established that the magnitude of repeated high peak pressures is of greater concern. Foot ulcers often occur in areas of callus. The callusing is not caused solely by plantar pressures but also from frictional shear forces.[71,113,173] Excessive shear and high peak plantar pressures have been implicated as causal agents in the formation of plantar foot ulcers in persons with diabetic neuropathy. Tissue breakdown occurs more rapidly when shear is increased.[72]

Brand[23,24] suggests that neuropathic ulcers result from repetitive trauma. He studied the histology of rat paws after repetitive stress and determined that an increase in temperature always occurred with repetitive stress. If the increase was followed immediately by a gradual return to normal, this change was thought to be caused by simple reactive hyperemia. If the temperature increase continued for 10 or more minutes, biopsies showed edema and collections of inflammatory cells. When the repetitions continued at a force equal to walking 7 miles per day at a fast pace on hard ground, epithelial hypertrophy with marked inflammation and necrosis of deeper tissues was noted at day 3. Ulceration occurred at day 10. When the same experiment was performed with 20% fewer daily repetitions and breaks on weekends, hypertrophy occurred without significant breakdown.[23,24] It can be inferred from this study that in patients with normal sensation, inflammation makes feet tender, and the tender spot is spared further stress until inflammation subsides. With a neuropathic foot, pain is no longer present to provide this feedback.

A study of the recurrence of neuropathic ulcerations revealed that nearly half of the recurrences were solely attributable to failure to follow recommendations to consistently offload.[82] Although a foot ulceration can be extensive, lack of subjective pain often leads to excessive weight-bearing. It is the medical team's responsibility to convey that pain sensation can be compensated by "intelligent anticipation."[24]

The risk of ulcers or amputations is increased in people who have diabetes for 10 or more years, are male, have poor glucose control, and have cardiovascular, retinal, or renal complications.[4] The cumulative risks of neuropathy, deformity, high plantar pressure, poor glucose control, and male sex are all additive factors for foot ulceration in persons with diabetes.

Neuropathy and peripheral arterial disease (PAD) are strong independent risk factors for the development of diabetic foot ulcers. Diabetic neuropathy and PAD are usually the major causative factors involved in diabetic foot ulcers. These two factors may act alone, together, or in combination with other conditions (microvascular disease, biomechanical abnormalities, limited joint mobility, and increased susceptibility to infection).[143,153] Diabetic patients are four times more likely to develop PAD and five times more likely to develop critical limb ischemia than the general population.[54] For every 1% increase in hemoglobin A1c, there is a corresponding 26% increased risk of PAD.[149] Key risk factors include the presence of peripheral neuropathy, foot deformity, PAD, ongoing or history of tobacco use,[171] or a history of foot ulceration or amputation.[88,103,119]

Evaluation of Neuropathic Ulcers

It is important to determine the etiology of the individual's peripheral neuropathy, assess causative and contributing factors of the ulcer, and review the health history to determine risk factors (diabetes, hypothyroid, alcohol dependency, vitamin D deficiency, metabolic diseases, pernicious anemia, advanced age). Evaluation of neurologic status in an at-risk foot should include a quantitative somatosensory threshold test. Light pressure assessment with the Semmes-Weinstein 5.07 (10-g) monofilament has been used in observational studies to support the association of loss of protective sensation and increased risk of ulceration.[19,38,118] Neuropathic patients should be examined under weight-bearing and non–weight-bearing conditions. Some deformities go unnoticed if a patient is not evaluated when weight-bearing. Patients' shoes should be evaluated for signs of abnormal, excessive, or irregular patterns of wear. Although some seek medical care because their neuropathy is painful, many patients with chronic painless sensorimotor peripheral neuropathy do not seek care until a significant problem (an ulcer) has developed.[20] These patients walk into clinic with no limp on an ulcerated foot. All too frequently, they have not been evaluated for peripheral neuropathy. Many are not aware they have a peripheral neuropathy or its associated risks.[5]

PAD is common in individuals with diabetes and should be assessed. Initial screening for PAD should include a history of claudication and an assessment of pedal pulses. Although clinical history and physical examination can be very suggestive of PAD, a definitive diagnosis should be established.[155] Early diagnosis, control of risk factors, and medical management as well as timely revascularization may aid in avoiding limb loss.

Neuropathic Ulcer Management

Five key elements to prevent foot problems include (1) identification of the at-risk foot; (2) regular inspection and examination of the at-risk foot; (3) patient, family, and health care provider education; (4) consistently wearing appropriate footwear; and (5) treatment of preulcerative findings.[29] To be successful, a diabetic foot care program needs to focus on prevention.[15,86,108] All individuals with diabetes and/or neuropathy should receive a careful examination of the feet at least once per year, noting deformities, signs of vascular compromise, loss of protective sensation, callus, and presence or absence of fungal infection in the web spaces.[27] Those with one or more high-risk foot conditions should be evaluated more frequently. The American Diabetes Association reports that comprehensive foot care programs can reduce lower extremity amputation rates among diabetic patients by as much as 45% to 60%.[46] It is thought that periodic examination of the feet (for sensory loss, foot deformity, improper shoes, and PAD) allows health care providers to recognize risk covariates for foot complications. The intent is to provide advice about preventive foot care (teaching and printed materials) and streamline triage for patients who require further evaluation.

The multidisciplinary team is optimal to achieve ulcer healing in the at-risk diabetic patient.[19,42,50,60,63] This team includes primary care physicians, nurses, podiatrists, and subspecialists (endocrinologists, rehabilitation medicine physicians, wound clinicians, orthopedic surgeons, vascular surgeons, interventional radiologists) who ideally provide coordinated care. Evidence supports the role of multidisciplinary programs in reducing diabetic foot ulcers and lower extremity amputation in at-risk individuals.[42,50,118] An integrated care pathway offers a potential tool to improve the standard of care for patients with neuropathic ulcers and/or critical limb ischemia. Decreasing the wait time to interventions may avert major amputation and reduce mortality. There are seven key elements for neuropathic ulcer treatment: (1) consistent offloading, (2) restoration of skin perfusion, (3) treatment of infection, (4) metabolic control and treatment of comorbidities, (5) local ulcer care, (6) patient and family education, and (7) preventing recurrence.[29]

In a patient with neuropathy, treat any preulcerative signs on the foot. This includes removing callus, protecting blisters or unroofing them if indicated, treating ingrown or thickened nails, and prescribing antifungal treatment for fungal infection.[29] Callus can be debrided with a scalpel by a foot care specialist or other health professional with experience and training in foot care. If callus is not removed, this will increase the forces and decrease the flexibility of the skin at a site already receiving excess force. Patient education to optimize foot care is advocated by most experts and clinical guidelines as an important strategy to prevent diabetic foot complications.[12,19,66,116] Surprisingly, the available body of evidence about the effectiveness of patient education programs has been inconclusive.[48,115,150] Patients at risk should understand the implications of the loss of protective sensation, the importance of foot monitoring on a daily basis, proper care of the foot including nail and skin care, and selection of appropriate footwear. Patients should be taught to substitute the loss of sensation with alternative senses (e.g., sight or touch) and avoid situations that may pose a risk of trauma (e.g., walking barefoot or using poorly fitted footwear).[12,19,66,116] Patients' understanding of these issues and their ability to conduct proper foot surveillance and care should be assessed. Swelling, redness, and increased warmth can be difficult for patients to monitor. Infrared thermography represents a potentially effective tool for detecting preulcerous states in diabetic foot patients.[58] A study evaluated home infrared temperature monitoring twice per day in diabetics at high risk for lower extremity ulceration and amputation. Study patients reduced their activity, contacted the study nurse when temperatures increased, and had significantly fewer foot complications.[105]

Offloading, or the relief of the mechanical load from the foot, is recommended as part of the standard treatment of diabetic foot ulcers.[34] Even with optimal local ulcer care, debridement, and dressings, an ulcer is unlikely to heal without consistent offloading.

Protective Provisional Footwear and Offloading. After any infection has been contained, edema controlled, and necessary revascularization performed, it is important to arrange for proper protective footwear for neuropathic or ischemic ulcers. Proper offloading and pressure reduction prevents further trauma and promotes healing. This is particularly important in the patient with decreased or absent sensation in the lower extremities. Patients should be counseled never to wear shoes that contribute to a foot ulcer.[33] It can be a challenge to convince a person who has not experienced foot discomfort to restrict footwear choices. The choice of offloading devices should be determined by the patient's physical characteristics, the patient's ability to use the device, and the location and severity of the ulcer.[66] This task can be best achieved with a comprehensive team.[37,52,84,91]

A walker or crutches can yield 20% offloading. Knee rolling walkers provide excellent offloading but need to be used with caution in those with neuropathy and risk of falling. Provisional footwear can be used for short-term offloading in patients with lower extremity ulcers. Provisional footwear protects the foot from pressure and shear; provides adequate length, width, and depth to accommodate the anatomic foot; accommodates edema/dressings; has a stiff gentle rocker sole to reduce peak pressure at the forefoot[85]; and has an ankle strap to avoid shearing. A 0.5-inch Plastazote insole is used for soft tissue supplementation. The provisional footwear should be replaced/repaired as needed. If a neuropathic ulcer persists despite offloading with a provisional shoe, total contact casting or an ankle-foot orthosis (AFO) should be considered. An AFO distributes weight-bearing forces over the entire lower leg to minimize peak pressures on the plantar surfaces. Many neuropathic deformities can be controlled only by crossing the ankle joint to reduce plantar pressures (Charcot arthropathy, hindfoot/ankle deformities). Total-contact casting has been shown to be beneficial for the treatment of neuropathic ulcers.[81,164] For patients with nonhealing neuropathic ulcers, total-contact casting is the gold standard. There is a gap between clinical practice for the treatment of neuropathic wounds and the International Working Group on the Diabetic Foot recommendations. Studies in Europe, the United States, and Australia have shown that practitioners underutilize total-contact casting, although the majority of them agree that these devices are the gold standard treatment.[69] The most compelling evidence that offloading accelerates ulcer healing comes from studies using total-contact casting for healing of noninfected neuropathic ulcers. Neuropathic ulcers that had not healed for many months or years typically heal in about 6 weeks in a total-contact cast.[3,123]

The total-contact cast redistributes weight-bearing forces, decreases edema, protects the ulcer and surrounding tissues, decreases sheer forces, localizes infection, protects the foot from outside contaminants, and provides immobilization of the ulcer and Charcot joint.[36,81] In previous research, casting allowed healing of even the most chronic neuropathic ulceration in an average of 33 to 38 days.[81,164] Total-contact casting and prefabricated cam walkers rendered irremovable have been shown to have a significantly higher proportion of healed diabetic neuropathic plantar forefoot ulcers when compared with removable offloading devices (including both cam walkers and footwear).[110,122] Although the total-contact cast is the gold standard, it cannot be used in the presence of infection or ischemia. In addition, patients may find it challenging to travel for the weekly appointments needed for cast application, bathe, or drive a car with a cast. The risk of inconsistent use of a removable device needs to be considered. When casting is not an option, we have had success with a custom solid ankle posterior shell polypropylene removable AFO with a built-in rocker sole and custom foot bed.[145] When a total-contact cast or other offloading provisional footwear is used, footwear on the sound side must be evaluated and adjusted if needed to avoid a leg-length discrepancy. After initial healing, using total-contact casting, a total-contact sandal with a custom foot bed of moderate density, closed cell, polyethylene foam and a Poron (PPT) interface can be fabricated. The patient wears this sandal for approximately 2 weeks until proceeding with definitive footwear.

Definitive Footwear. Therapeutic (prescription) footwear decreases weight-bearing pressure and shear forces applied to the foot.[140] Therapeutic footwear is used for long-term redistribution of weight-bearing forces after ulcer healing. Therapeutic footwear should not be used for ulcer healing.[8] Proper therapeutic footwear is the basis of long-term neuropathic foot management. Patients with neuropathy should be advised to break in new shoes gradually to minimize the formation of blisters and ulcers. Conservative pedorthic management (footwear, shoe modifications, custom foot orthosis, and partial foot prostheses) of the diabetic foot has been shown to be an effective method to prevent ulcers, amputations, and reamputations. Pressure relief and load redistribution are better achieved with custom total-contact foot orthoses than off-the-shelf insoles in patients with diabetes, neuropathy, and foot deformity.[30] Patients with evidence of increased plantar pressure (erythema, increased warmth, callus, or measured pressure) should use footwear that cushions and redistributes the pressure. Protective footwear should be prescribed for any patient at risk for amputation (PAD, neuropathy, previous amputation, previous ulcer, periulcerative callus, foot deformity, and evidence of callous formation).

Proper shoe fit is critical. Either a loose shoe or a tight shoe has the potential to increase shear, friction, and/or pressure on the foot. Improper footwear is a common cause of diabetic ulcers.[144] Therapeutic footwear can help reduce foot ulcers.[52,146] The shoe must be the correct shape and provide adequate depth to properly accommodate the patient's foot. The shoe should not cause areas of point pressure across the dorsum or mediolateral foot. Shoes with laces or hook-and-loop closure systems tend to provide better fit than slip-on shoes. Persons with neuropathy should avoid slip-on shoes. A shoe should be constructed of soft leather without seams over bony prominences. Diabetic footwear prescriptions should include extra-depth shoes. An extra-depth shoe is an oxford-type or athletic shoe with an additional ¼ to ⅜ inch of depth throughout the shoe.[89,90] The additional volume provided in an extra-depth shoe makes it ideal for patients with diabetes. An extra-depth shoe can accept a foot orthosis or AFO without affecting the fit of the shoe and accommodate deformities such as hammertoes. These shoes offer even more space when the removable factory inlays are taken out. The sole of the shoe should be shock absorbing and easily modifiable to accommodate deformities. The most common sole modification is the rigid rocker sole, which allows ambulation with reduced pressure on the forefoot and toes. This modification requires an addition to the sole thickness with an apex at 1 cm proximal to the metatarsal head for the foot to roll over the forefoot. The length of the shoe must be ½ to ¾ inch beyond the longest toe to accommodate the natural elongation of the foot during ambulation. The arch length is measured at the metatarsal heads and ensures that the shoe bends at the same location as the foot bending motion. The shoe and insole must be replaced when wear prevents the optimal function of either.

When an off-the-shelf shoe does not provide appropriate fit (because of deformity or variance in size from left foot to right), it may be necessary to modify the shoe or provide a custom-made shoe. Patients with foot deformities that cannot be accommodated by standard therapeutic footwear should have custom shoes that provide appropriate fit, depth, and a rocker sole. Custom-made shoes are fabricated by creating a positive model from a mold of the patient's foot. The shoe is then constructed around this model. Since custom shoes are made directly from molds of the feet, they offer the best possible accommodation and protection but can be quite expensive and lack the cosmetic appeal of commercially available shoes. The plantar pressure gradient is important.[124] Using a foot pressure measurement system with pressure-sensitive insoles, plantar pressures and their distribution may be measured with the patient wearing the shoes and foot orthoses. Reduction of peak plantar pressures often focuses on the forefoot as diabetic ulcers occur most frequently in this area.[124,172] Research has looked at the reduction of forefoot pressures using shoes, rocker soles, and various types of foot orthoses.[26,80,106,128,159] Although much has been written about areas of high peak pressures as a predictor of foot ulcers, research has shown that there is not a particularly high correlation between the two. [9,104,163,172]

Although reducing elevated pressure is important, reducing the duration of maximum pressure and shear stresses may be even more important.[41] A polytetrafluoroethylene material called ShearBan is widely available to the orthotic, prosthetic, and pedorthic industry to decrease shear. The self-adherent, heat-moldable material can be used virtually anywhere (inside a shoe, brace, or prosthetic socket).

Dressings. There is no specific dressing for neuropathic ulcers. Assessing the ulcer characteristics will determine the dressing choice. For example, an absorbent dressing may be beneficial in the presence of moderate to heavy drainage. Early advanced ulcer care products may be more cost effective than standard ulcer care products to decrease the incidence of lower extremity amputations. Given that over 80% of chronic wounds will have an associated biofilm, consideration should be given to strategies that mitigate biofilm and reduce the associated inflammation. Adjunctive strategies such as moist compress applications of hypochlorous acid (pH of 5.5) for 10 minutes at dressing changes (ideally initially daily to decrease bioburden) can decrease critical colonization of wounds and improve wound outcomes.[79,132]

Biofilm

In chronic ulcers, bacteria form a protective biofilm that goes unrecognized by host cells and is impermeable to most systemic and topical antimicrobials. A biofilm develops when microorganisms adhere to and begin to grow on the surface of an ulcer. Bacteria within a biofilm secrete a gluelike matrix.[167] This matrix allows the biofilm to establish itself more completely on the ulcer and makes it more resistant to antimicrobial treatment and the patients' immune system.[139,167] Once reaching this stage, the microbes within a biofilm are difficult to kill. It is therefore important to prevent biofilm formation.[139] Biofilms adversely affect ulcer healing because they impair the immune response, delay epithelialization, and decrease the growth of granulation tissue.[139] Biofilms are more common in chronic ulcers, leading ulcer healing to become stagnant with ulcers in an inflammatory state.[167]

Antibiofilm strategies differ from standard care. Although ulcer debridement is essential to disrupt this biofilm and allow penetration of host cells and dressing materials, a multifactorial approach is necessary because biofilms are polymicrobial and may reestablish themselves within 24 hours after debridement.[167] In addition to debridement and antibiotics, topical antiseptics such as cadexomer iodine preparations have been helpful to prevent biofilm formation.[167]

Pressure Injury

Pressure injuries are defined as "localized damage to the skin and/or underlying tissue, as a result of pressure or pressure in combination with shear. Pressure injuries usually occur over a bony prominence but may also be related to a medical device or other object."[10] *Decubitus ulcer* is the oldest term used to describe pressure injuries; it dates to 1777. This term specifically relates to dead tissue caused from lying down for long periods of time. The term *bedsore* arose in 1976 and again alludes to the fact that the ulcers are caused from lying in bed.[10] It is now recognized that pressure injuries do not only occur from patients lying down.[10] *Pressure sores/ulcers* were the terms that were most commonly used in the 1980s and 1990s.[10] These changes in terminology are more generic and do not confine the cause to being in bed.[10] In 2016, the National Pressure Ulcer Advisory Panel held a consensus conference and confirmed that the new term would be *pressure injury*, and transformed their name to the National Pressure Injury Advisory Panel (NPIAP). This terminology change is more inclusive because pressure injuries present as both intact and ulcerated skin.[10] Therefore *pressure injury* is the current nationally recognized and preferred term. An injury occurs as a "result of intense and/or prolonged pressure or pressure in combination with shear. The tolerance of soft tissue for pressure and shear may also be affected by microclimate, nutrition, perfusion, co-morbidities and condition of the soft tissue."[10]

Pressure injuries contribute to pain, suffering, decreased quality of life, and increased health care costs. People of all ages, from the very young to the very old, can develop pressure injuries. Identification of at-risk individuals, prevention approaches, and treatment plans require multidisciplinary collaboration.

In 2008, the Centers for Medicare & Medicaid Services (CMS) stopped paying the additional costs associated with hospital-acquired pressure injuries.[11] At that time, hospitals stepped up prevention efforts to reduce the incidence of pressure injuries. It is estimated that the cost of treating pressure injuries is up to $152,000 per injury in the United States.[10] The global prevalence of pressure injuries was evaluated in a meta-analysis looking at literature from 2008 to 2018. Li et al.111 reported:

> *Of 7,489 studies identified, 42 were included in the systematic review and 39 of them were eligible for meta-analysis, with a total sample of 2,579,049 patients. The pooled prevalence of 1,366,848 patients was 12.8% (95% CI 11.8-13.9%); pooled incidence rate of 681,885 patients was 5.4 per 10,000 patient-days (95% CI 3.4-7.8) and pooled hospital-acquired pressure injuries rate of 1,893,593 was 8.4% (95% CI 7.6-9.3%). Stages were reported in 16 studies (132,530 patients with 12,041 pressure injuries). The most frequently occurring stages were Stage I (43.5%) and Stage II (28.0%). The most affected body sites were sacrum, heels and hips. Significant heterogeneity was noted across some geographic regions. Meta-regression showed that the year of data collection, mean age and gender were independent predictors, explaining 67% variability in the prevalence of pressure injuries. The year of data collection and age alone explained 93% of variability in hospital-acquired pressure injuries rate.*

Sugathapala et al. reviewed the prevalence of pressure injuries in nursing homes and found it to be similar to that found in the acute hospital setting.[157]

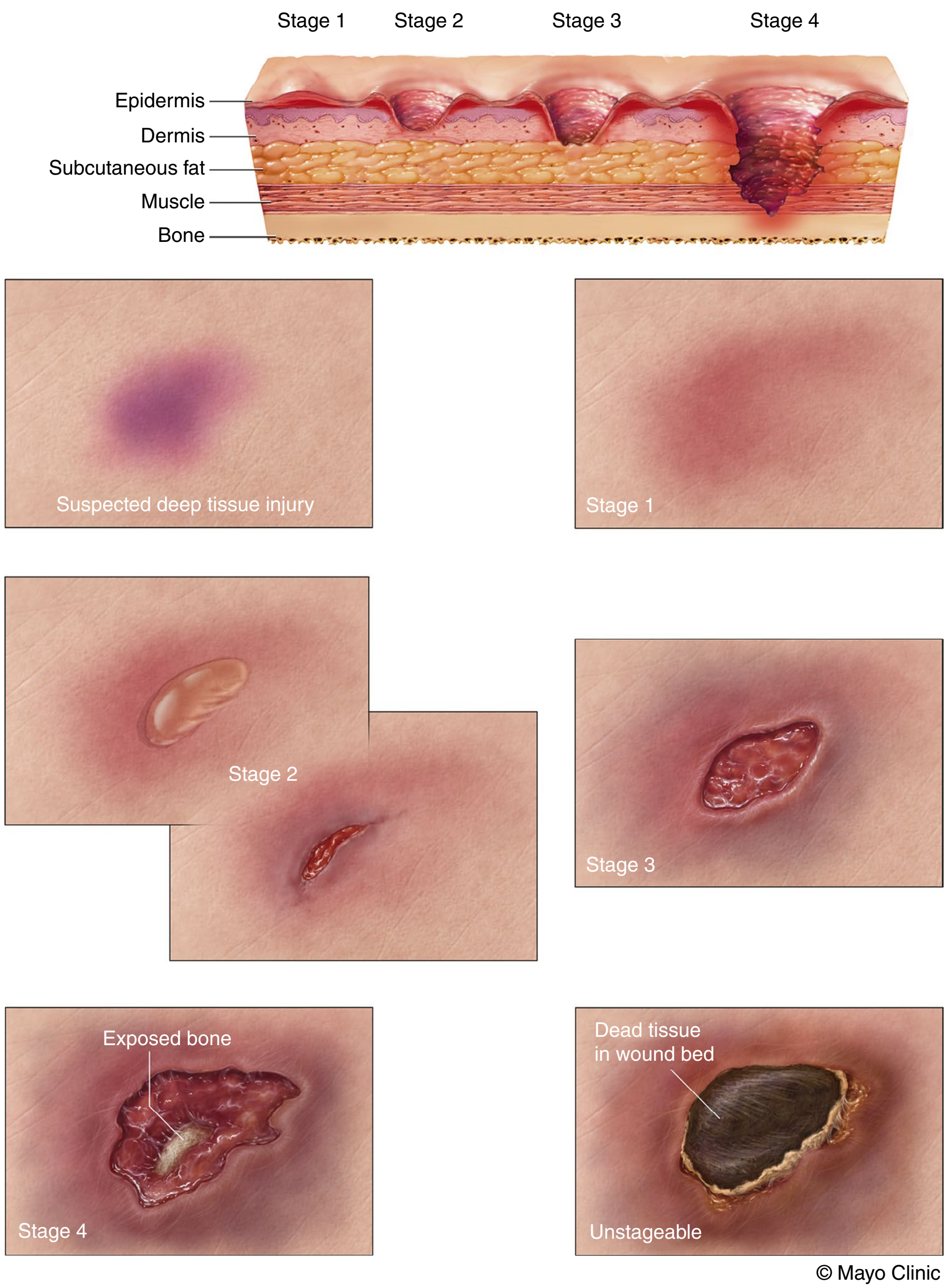
Documentation

As previously mentioned, in 2008 CMS stopped paying the additional costs associated with hospital-acquired pressure injuries.[11] CMS determines whether a pressure injury is hospital acquired or present on admission based on prescriber documentation. It is imperative that prescribers and nursing communicate and reach agreement on whether a pressure injury was present on admission, the stage of the pressure injury, and the treatment plan.

Pressure Injury Staging

Pressure injury staging involves multiple factors (Fig. 25.5).[10]

Stage 1 Pressure Injury: Nonblanchable Erythema of Intact Skin. Intact skin with a localized area of nonblanchable erythema may appear differently in darkly pigmented skin (Fig. 25.6). The presence of blanchable erythema or changes in sensation, temperature, or firmness may precede visual changes. Color changes do not include purple or maroon discoloration; these may indicate deep tissue pressure injury (DTPI).



• **Fig. 25.5** Stages of pressure injuries. (From Mayo Clinic Patient Education. *Stages of Pressure Injuries (MC5896-02)*. Mayo Clinic; 2017.)

• **Fig. 25.6** Stage 1 pressure injury.

Stage 2 Pressure Injury: Partial-Thickness Skin Loss with Exposed Dermis. Partial-thickness loss of skin has exposed dermis. The ulcer bed is viable, pink or red, moist, and may present as an intact or ruptured serum-filled blister (Fig. 25.7). Adipose (fat) is not visible, and deeper tissues are not visible. Granulation tissue, slough, and eschar are not present. These injuries commonly result from adverse microclimate and shear in the skin over the pelvis and shear in the heel. This stage should not be used to describe moisture-associated skin damage (MASD), including incontinence-associated dermatitis, intertriginous dermatitis, medical adhesive–related skin injury, or traumatic ulcers (skin tears, burns, abrasions).

• **Fig. 25.7** Stage 2 pressure injury.

• **Fig. 25.9** Unstageable.

• **Fig. 25.8** Stage 4 pressure injury.

Stage 3 Pressure Injury: Full-Thickness Skin Loss. Full-thickness loss of skin occurs, in which adipose (fat) is visible in the ulcer and granulation tissue and epibole (rolled ulcer edges) are often present. Slough and/or eschar may be visible. The depth of tissue damage varies by anatomic location; areas of significant adiposity can develop deep ulcers. Undermining and tunneling may occur. Fascia, muscle, tendon, ligament, cartilage, and/or bone are not exposed. If slough or eschar obscures the extent of tissue loss, this is an unstageable pressure injury.

Stage 4 Pressure Injury: Full-Thickness Skin and Tissue Loss. Full-thickness skin and tissue loss occurs with exposed or directly palpable fascia, muscle, tendon, ligament, cartilage, or bone in the ulcer (Fig. 25.8). Slough and/or eschar may be visible. Epibole (rolled edges), undermining, and/or tunneling often occur. Depth varies by anatomic location. If slough or eschar obscures the extent of tissue loss, this is an unstageable pressure injury.

At times it can be difficult to distinguish a partial-thickness ulcer from a full-thickness ulcer. A partial-thickness ulcer involves only the epidermis and dermis, whereas a full-thickness ulcer extends through the epidermis and dermis to the subcutaneous tissue as well as muscle, bone, and fascia. A partial-thickness ulcer will heal through reepithelization and can heal from the center of the ulcer to the edges, whereas a full-thickness ulcer heals through granulation, reepithelization, and contraction.[28] A partial-thickness ulcer will often have intact hair follicles present in the ulcer bed, whereas a full-thickness ulcer will not.[10] Also, when the surrounding skin is shifted and moved, a partial-thickness ulcer will move with the surrounding skin. The base of a full-thickness ulcer will stay stationary when the surrounding skin is shifted.

Unstageable Pressure Injury: Obscured Full-Thickness Skin and Tissue Loss. Full-thickness skin and tissue loss occurs in which the extent of tissue damage within the ulcer cannot be confirmed because it is obscured by slough or eschar (Fig. 25.9). If slough or eschar is removed, a stage 3 or 4 pressure injury will be revealed. Stable eschar (i.e., dry, adherent, and intact without erythema or fluctuance) on the heel or ischemic limb should not be softened or removed.

Deep Tissue Pressure Injury: Persistent Nonblanchable Deep Red, Maroon, or Purple Discoloration. Intact or nonintact skin with localized area of persistent nonblanchable deep red, maroon, purple discoloration or epidermal separation reveals a dark ulcer bed or blood-filled blister. Pain and temperature change often precede skin color changes. Discoloration may appear differently in darkly pigmented skin. This injury results from intense and/or prolonged pressure and shear forces at the bone-muscle interface. The ulcer may evolve rapidly to reveal the actual extent of tissue injury or may resolve without tissue loss. If necrotic tissues, subcutaneous tissue, granulation tissue, fascia, muscle, or other underlying structures are visible, this indicates a full-thickness pressure injury (unstageable, stage 3, or stage 4). Do not use DTPI to describe vascular, traumatic, neuropathic, or dermatologic conditions. The NPIAP[10] has also defined medical device–related pressure injuries and mucosal membrane pressure injuries.

Medical Device–Related Pressure Injury. Medical device–related pressure injuries result from the use of devices designed and applied for diagnostic or therapeutic purposes. The resultant pressure injury generally conforms to the pattern or shape of the device. The injury should be staged using the staging system.

Mucosal Membrane Pressure Injury. Mucosal membrane pressure injury is found on mucous membranes with a history of a medical device in use at the location of the injury. Because of the anatomy of the tissue, these pressure injuries cannot be staged.

Pressure Injury Prevention

Pressure injuries are caused by a mechanical load, which is sustained and applied to tissues causing deformation, usually over a bony prominence. They can also be located where devices come in contact with tissues. Tissue damage depends on the weight of the mechanical load as well as the duration of time. A low mechanical load applied for an extended time frame as well as a high load for a brief time frame can both cause tissue deformation and damage.[127] The evidence demonstrates that deformation of the tissues can result in two types of damage. First, ischemic injury occurs with actual occlusion of blood supply leading to hypoxia, blocking of nutrient supply, and interfering with the removal of waste products. Upon reperfusion after prolonged ischemia, oxygen-free radicals are released that can cause further tissue damage. Second, it is hypothesized that direct tissue deformation causes a rupture of the cytoskeleton and stretches the plasma membrane, which eventually leads to cell death. Muscle and fat tissues are most susceptible to this force. The layers of skin are stiffer, making tissue deformation less likely to occur.[127]

When a patient is lying in bed and the head of the bed is raised, gravitational forces pull the body down while friction on the bed's surface tries to hold it in place.[28] These opposing forces cause the blood vessels in the deeper tissue layers to stretch and bend, impeding circulation.[28] It can occur even when an individual is lying on a flat surface.[127] Many hospitals have well-defined guidelines and protocols in place for pressure injury prevention. It is important to consider the patient's risks outside the hospital room (e.g., transport stretchers, operating tables, radiograph or magnetic resonance imaging tables, recliners used during hemodialysis). Successful prevention of pressure injuries involves a multidisciplinary approach and good communication. Nursing, medical and surgical teams, transport staff, imaging and procedural staff, rehabilitation providers, and dieticians are all critical to prevention. Physical and occupational therapists have the knowledge and skills to assist with equipment needs, such as wheelchair cushions. Pressure injuries do not occur only in hospitals. Assessment of the patient's environment, daily activities, behaviors, and support can help to determine protective measures concerning awareness of potential risks outside the hospital. Protective measures may include a cushion for a chair, a foam dressing to protect the bridge of the nose when initiating a respiratory mask, or ensuring that shoes fit appropriately. Individualized care of each patient should include a careful assessment, identification of specific risk factors, and implementation of targeted strategies for pressure injury risk reduction. Patient and caregiver education is a crucial part of the plan.

Risk Assessment

Factors that increase the risk for pressure injuries include reduced mobility, malnutrition, friction, microclimate, and physiologic parameters such as anemia, use of vasopressors, and vascular disease. There are standardized tools for determining the level of risk for pressure injury development in adult and pediatric patients. The screening tools generally assess sensory perception, skin moisture, patient activity level, patient mobility, patient nutrition, and friction or shear. Some common risk scales are the Braden[22] (Table 25.1), Braden Q, Norton Scale,[76] Braden QD,[40] and neonatal skin score. Knowledge of an individual's risk areas can be utilized to customize the prevention plan. The use of an assessment scale and attention to physiologic factors provide a standard way to predict risk and can help guide interventions.[127] Minimizing the risk factors is key; we may not be able to eliminate them. Be aware of the risk factors and make attempts to reduce the effects. Risk status for a patient should be communicated across the continuum, including when the patient has tests and procedures. Although risk assessments are historically looked upon as beneficial and necessary, Padula et al.[135] recently looked at cost effectiveness of preventive interventions and found that prevention for all patients, regardless of how they rate on the risk assessment tool, is cost effective in greater than 99% of the probabilistic simulations they ran. Therefore providing prevention interventions for all hospitalized patients may be on the horizon.

Risk Factors

When patients are sedentary or bedfast, they should be considered at risk for developing pressure injuries.[127] A painful stimulus in a sensate patient will cause that person to reposition the body part and offload the pressure. When neurologic impairment is present, the pain stimulus will not be recognized and therefore the tissue may not be offloaded. Friction plays a role in skin damage and can lead to increased internal tissue shear and strain with pressure

TABLE 25.1 Braden Headings

Sensory/Perception	Moisture	Activity	Mobility	Nutrition	Friction and Shear
1 Completely limited	1 Constantly moist	1 Bedfast	1 Completely immobile	1 Very poor	1 Problem
2 Very limited	2 Often moist	2 Chairfast	2 Very limited	2 Probably inadequate Probably inadequate	2 Potential problem
3 Slightly limited	3 Occasionally moist	3 Walks occasionally	3 Slightly limited	3 Adequate	3 No apparent problem
4 No impairment	4 Rarely moist	4 Walks frequently	4 No limitations	4 Excellent	

See Braden and Bergstrom[22] for details of the Braden Risk Assessment Scale. Total Braden score 15–16 mild risk; 12–14 moderate risk; less than 12 high risk; 15–18 is considered mild risk for those greater than 75 years. Modified from Braden BJ, Bergstrom N. Predictive validity of the Braden scale for pressure sore risk in a nursing home population. *Res Nurs Health.* 1994;17(6):459-470.

injury development, but friction in and of itself only causes superficial skin damage. If an ulcer is primarily caused by friction, it should be called a friction injury. Microclimate is the degree of humidity and temperature that the skin is exposed to when resting on a support surface.[127] Excessive moisture and heat increase the risk for skin breakdown.[127] It is important to find the appropriate balance between skin moisture and dryness because skin that is too dry lacks suppleness and cracks more easily.[127]

When a person has had a previous pressure injury, the healed area will never have the tensile strength that it had before the original injury, making the area more susceptible to future injuries and breakdown.

Medical devices have also been implicated in pressure injury development, including cervical collars, casts, respiratory equipment (including oxygen tubing and noninvasive positive pressure airway devices), compression wraps, nasogastric tubes, and endotracheal tubes.[53,127] The presence of edema and moisture has the potential to increase the risk for pressure injury development under medical devices.[53,127] Patients may complain of pain, burning, or tingling at the site or have no symptoms of discomfort. The device should be removed at regular intervals so skin inspection can occur whenever possible. It is important for the provider to specify when a device such as a cervical collar should not be removed. When providers are changing a device or dressing, such as an orthopedic splint during a procedure, they need to assess the skin for any signs of pressure injury before replacing the device.[127] Tubes should be repositioned, and five-layer prophylactic foam or antishear dressings may be used to prevent pressure injuries.[127] Assuring that medical devices fit properly is an important prevention strategy.[127]

Prevention Strategies

Sensory Perception Deficit Recognition. Care providers need to be cognizant of deficits in sensory perception to proactively reduce areas of pressure. Persons who are comatose or anesthetized and those with hemiparesis, tetraplegia, paraplegia, and peripheral neuropathy are examples of high-risk populations. Repositioning and heel elevation with boots are important prevention interventions for patients with sensory deficits.

Activity and Mobility Interventions for Pressure Redistribution. Efforts to increase activity and mobility aim to decrease the duration and intensity of pressure. Common interventions include getting people out of bed as much as possible, turning to a new position, positioning at 15- to 30-degree angles when side lying to avoid excessive pressure on the trochanter, shifting positions when seated, and using pressure-relieving cushions. Prophylactic use of foam dressings over the sacral area of patients has resulted in a decrease in the number of hospital-acquired pressure injuries. The head of the bed should be at or below 30 degrees when possible to avoid additional pressure to the coccyx and sacral area.[61,127]

Support Surfaces. Support surfaces can help manage microclimate and pressure redistribution and are essential for preventing and treating pressure injuries. However, nursing interventions, such as repositioning and heel elevation, are still necessary. Mattresses can be purchased or rented depending on what properties are desired. Some of these properties include immersion, alternating pressure, low air loss, pulsation, and different types of foam or gel. When exploring what surface will work well for an acute care setting, it is integral to consider the differences in patient population, such as general care vs. the intensive care unit (ICU), and ensure that the surface will meet the needs. The bed frame and functionality are also important to take into consideration to ensure it will meet the needs of the patients and staff. Evidence shows that use of "smart beds," which provide real-time pressure mapping display and automatic bed adjustment based on pressure and time, reduced pressure injury in an ICU setting.[14] More recent studies related to continuous pressure mapping have shown promise in prevention of pressure injuries but continue to require direct interaction by nursing staff.[14] There is also a large variety of chair cushions available, and it is important to establish criteria and directions for use. Chair cushions are typically made of static air, gel, or foam. Donut-shaped chair cushions should be avoided because they can restrict blood flow to the skin in the middle of the donut. Pressure redistribution surfaces can be found in both bed and chair variations. Despite the type of support surface, it is critical to follow regimented turning and positioning schedules and adjust frequency based on patient assessment. The CMS has information for surface options, required indications, and necessary documents when ordering for home use.[117]

Nutritional Support. Calories need to be adequate to maintain nutritional status. Caloric needs must be met first for protein to be used as a contributor to the ulcer healing process rather than as an energy source. Biochemical data such as albumin and prealbumin are not good indicators of nutritional status. Quickly identifying and treating undernutrition will assist in both prevention and management of pressure injury risk and healing. Low body mass index puts a patient at higher risk for pressure-related injuries.[127] The NPIAP[127] recommends the following nutritional guidelines to support ulcer healing: (1) screen and assess nutritional status on admission and with each change in condition; (2) provide adequate calories, protein, vitamins, and minerals; and (3) maintain hydration by providing sufficient fluid intake. The nutritional supplements that are used most often in therapeutic regimens are vitamins A, B complex, C, D, E, and zinc.[138,141,151] If a nutrient deficiency is suspected, determining the blood level will assist in proper identification and adequate replacement of micronutrients to facilitate wound healing. Micronutrients are an important aspect of endothelial cell function and wound healing.[83] Nutritional and/or endocrine consultations may be of great benefit to provide proper micronutrient dosing regimens in complex patients with multiple comorbidities.

Moisture Management. Moisture in combination with pressure or pressure with shear can lead to pressure injury.[127] MASD can occur from incontinence, drainage from ulcers or tubes, or sweating. When exposed to moisture, the skin should be cleansed with a pH-appropriate cleanser (pH ≤ 6)[147] to help maintain the acid mantle protection (natural skin pH 5).[16] Barrier sprays and creams can be used to reduce the skin's exposure to extremes of moisture and dehydration. Absorbent products are available that wick moisture away from the skin. Foam dressings composed of polyurethane help to absorb moisture.[127] One management strategy is to limit the use of disposable briefs that trap heat and moisture on the skin. Disposable underpads are being made with absorbent polymers, making them more effective than plastic-backed underpads.[127]

Shear and Friction Reduction. Reducing shear can be accomplished by thoughtful positioning; for example, raising the knees before raising the head of the bed will reduce the shear to the coccyx and sacral area. There are antishear textiles available, including linens and underpads.[127] Early studies of silk linens or undergarments show potential for stage 1 and 2 pressure injury prevention, but further studies are needed.[127]

Other

It is important to be aware that there are many other causes of ulcers, including but not limited to trauma, vasculitis/vasculopathy, neoplasm (higher incidence in the extremities of patients with chronic lymphedema,[6] including squamous cell, basal cell, and angiosarcoma), connective tissue disease, medication related, calciphylaxis,[55,67,98] and pyoderma gangrenosum.[2,156] A differential diagnosis of the ulcer etiology can be aided by considering history, location, shape, depth, edges, and appearance.[17,49,114] Dermatologists are a critical part of diagnostic determination in these atypical ulcerations. Performing a biopsy that includes dermatologic pathology and atypical pathogen cultures (fungal, acid-fast bacilli), especially in an immunocompromised patient (posttransplantation, HIV/AIDS, on chronic immunosuppressive biologics), can be an important diagnostic procedure to help determine the underlying etiology of nonhealing ulcers.[43]

Ulcer Management

The main objectives of ulcer care are to reduce or eliminate causative factors, including pressure and shear; provide systemic support by managing comorbidities, such as malnutrition, hyperglycemia, and hypotension; and create a physiologic environment conducive to ulcer healing.[28] Basic concepts are summarized in the following anonymous quote: "If it's dirty, clean it. If it's deep, fill it. If it's open, cover it. If it's dry, moisten it. If it's wet, absorb it." (Refer to sections on debridement, biofilm, and dressings for further guidance.) Ulcer care principles include preventing infection; cleansing ulcers to reduce bioburden; removal of devitalized tissue; filling dead spaces, such as tunneling; optimizing the ulcer environment to promote granulation and reepithelialization; avoiding maceration, trauma, friction, or shearing forces; relieving pressure; and evaluating for reversible underlying conditions that may predispose to ulcer development or impede ulcer healing and pain management.

Bacterial Colonization Versus Infection

All chronic ulcers contain bacteria, and all ulcers are generally contaminated within 48 hours.[152] Contaminated ulcers have a low number of bacteria that do not replicate. Colonized ulcers have bacteria adherent to the ulcer surface, which replicate and form colonies without causing a negative reaction. When an ulcer is critically colonized, a significant number of bacteria form a biofilm, which delays ulcer healing. Infected ulcers result after bacteria have invaded the tissue, reproduce, and cause a host reaction, which is generally associated with fever, warmth, swelling, pain, erythema, purulent drainage, friable granulation tissue, and ulcer degradation. When the bacterial count progresses to infection, ulcer healing is impaired because of the metabolic load inflicted by bacteria. Nonviable tissue inhibits cells needed to stimulate granulation tissue. Significant bacterial bioburden overwhelms healing and prolongs the inflammatory phase. Edema and exudate prevent cell migration and prevent leukocytes from reaching the ulcer. Once these factors are controlled, ulcer healing can progress. Research has shown that when bioburden reaches levels greater than 10^5 bacteria per gram of tissue, this often results in ulcer infection.[78] Research also suggests that low levels of bacteria may aid ulcer healing by producing proteolytic enzymes that help ulcer debridement and stimulate proteases from neutrophils.[134]

Debridement

Simple sharp debridement using topical lidocaine and occasionally surgical debridement might be necessary. Necrotic debris not only causes a foreign body reaction but also provides an ideal environment for microorganisms. Dead tissue, whether black and dry or yellow and fibrinous, also prevents proper assessment of the ulcer depth. Ulcers with a small amount of superficial necrotic tissue can be mechanically debrided, gently, using saline-moistened coarse mesh gauze. Enzymatic agents cannot remove a hardened, black eschar or a large amount of necrotic tissue; however, they can loosen the eschar to facilitate sharp debridement. Pulsed-water debridement can be a useful adjunct for ongoing debridement of large ulcers (Table 25.2). Indications for surgical debridement include the presence of a draining sinus that communicates with a foreign body (orthopedic hardware, exposed mesh or other surgical repair material, fragmenting bone and osteomyelitis), an infected nongranulating ulcer, a large abscess, necrotizing fasciitis, and exposed nongranulating cartilage.[25] In a patient with a chronic ulcer, the diagnosis of osteomyelitis can be clinically challenging. Although bone scans are used to diagnose osteomyelitis, magnetic resonance imaging is more sensitive for detecting the early bone changes of osteomyelitis.[25] X-rays typically do not demonstrate classic signs of osteomyelitis until advanced stages. Once the presence of osteomyelitis is established, the area of infected bone should be debrided or excised. Caution must be used to limit the debridement to devitalized tissue. It is important to evaluate the arterial supply carefully before sharp debridement. If the blood supply is inadequate, the margin of debridement is at risk for necrosis. With distal ischemia (dry gangrene) in a patient who is not a candidate for revascularization, management can include protection of the extremity while awaiting demarcation and subsequent autoamputation in the affected region. Autoamputation is used to preserve as much of the residual limb as possible.

Considerations When Selecting Dressings

The challenge for clinicians is to understand the role and benefit of ulcer products and when they are appropriate for use. Novel ulcer care therapies have been intensely researched; however, many of the studies show conflicting evidence, resulting in an assortment of available products but few RCTs to support their use. In the 1960s, the concept of moist ulcer healing was introduced.[152] In the 1990s, advanced ulcer products became commercially available, and in the early part of this century, bioengineered cellular and tissue products became available. Despite these advances in ulcer products, understanding the mechanisms of normal and abnormal ulcer healing remains fundamental. Before initiating any ulcer dressing, the ulcer characteristics must be addressed. First, the ulcer must be suitable for healing. All ulcer dressings have the ability to manipulate one or more aspects of the healing process, and deciding which product to use depends on aligning ulcer characteristics with treatment capabilities. The importance of recognizing the ulcer type, the need for debridement, amount of exudate, and presence of infection cannot be ignored. In most cases, healing and prevention of further injury are the goals. However, for palliative patients, the focus of care may be on reducing/eliminating pain, controlling odor (metronidazole cream or crushed tablets as a powder),[1] and preventing infection, not necessarily on ulcer healing. In these situations, it is important to allow for an environment that promotes healing but does not supersede comfort and quality of life.

TABLE 25.2 **Types of Debridement**

Types of Debridement	Properties	Things to Consider
Autolytic	Dressings that provide a moist environment that allow slough and eschar to soften and then be more easily removed from the ulcer bed	Slower process Low cost and selective
Chemical/enzymatic	Collagenase selectively debrides necrotic tissue while safe to clean tissue[142]	Some products are more selective or gentler than others Collagenase requires a moist ulcer surface[142] Ulcer cleansers that change the pH of the ulcer may hinder enzymatic activity[142]
Conservative sharp	Can sharply remove nonviable tissue as long as the patient can tolerate and injury to healing tissue can be avoided	Can be done at the bedside by trained individuals
Bio active/larval	Creates an alkaline environment that inhibits bacterial growth Selective but can be distasteful for some patients	Maggots Challenging to contain the insects
Noncontact, low-frequency ultrasound	Cavitation and streaming breaks up bacterial cells, reduces bioburden, provokes a cellular response that seems to promote ulcer healing[45]	
Pulsatile lavage	Uses ulcer irrigation and suction to remove tissue and bacteria	
Surgical sharp debridement	Selective, useful when there are large areas of necrotic or nonviable tissue	Remove necrotic or nonviable tissue that is impeding healing or is infected
Mechanical	Nonselective. A dry gauze dressing or textured polyester pad provides mechanical debridement	Useful with heavily necrotic ulcers and no viable tissue at the surface that can be damaged

Data from Brodsky JW, Schneidler C. Diabetic foot infections. *Orthop Clin North Am.* 1991;22:473-489; Kavros SJ, Liedl DA, Boon AJ, et al. Expedited wound healing with noncontact, low-frequency ultrasound therapy in chronic wounds: a retrospective analysis. *Adv Skin Wound Care.* 2008;21(9):416-423.

Wound Care Products

Many substances have been used over time to treat ulcers in an attempt to stop bleeding, absorb drainage, and promote healing. Some of these materials included honey, animal fat, cobwebs, mud, leaves, sphagnum moss, or animal dung.[62] Although many of these natural substances would later prove to be of little benefit, others, such as honey, have been studied and show some value. Early ulcer dressings were simple and focused mainly on product composition. Currently, the biomaterials used in wound dressings are diverse. Some products keep the native composition of the extracellular matrix and contain growth factors and cytokines that modulate the inflammation process and promote wound healing, whereas some acellular dermal matrices trap and bind the recipient's own cells and growth factors, which support neovascularization and tissue regeneration.[35] As more sophisticated dressings are developed, new standards are required to prove that these products performed a specific function in consistent and reproducible ways. Few high-quality RCTs evaluating ulcer dressings exist, and these do not show superiority of one dressing over another. Using clearly defined processes to study patient-centered outcomes will help clinicians decide whether a specific treatment will work and whether the clinical benefits outweigh the potential harm or costs. Clinically, ulcer care should be evidence based, but there is a misperception regarding the various approaches to ulcer management. Equipped with the knowledge of the different dressing materials presented in this chapter, effective ulcer care strategies can be formulated.

Each facility has different dressing availability. Hospital settings will most likely have more limited options than outpatient ulcer healing clinics. The types of dressings and their general use are reviewed in Table 25.3. Always read the manufacturers' inserts before using a product to ensure appropriate use of the product for your patient. Check for allergies and other contraindications. Ulcers heal best in a moist, clean environment. This allows for growth factors and epithelial cells to facilitate ulcer edge contraction and healing. The faster rate of healing observed in ulcers that are kept moist has been attributed to the avoidance of tissue injury during dressing removal and the fact that desiccation of the ulcer inhibits epithelialization. It has been recently discovered that moisture-retentive dressings also facilitate dissolution of necrotic and fibrinous tissue (autolysis and fibrinolysis). Dressings that promote lysis of necrotic tissue and fibrin can facilitate the release of growth factors and reduce time to healing.[97] Dressings can provide painless debridement, stimulate granulation tissue, and encourage epithelialization.

The main principles for dressing selection include keeping the ulcer bed moist and addressing the following: bacterial bioburden, the nature and volume of exudate, condition of ulcer bed tissue, condition of surrounding skin, ulcer size, depth, location, presence of tunneling or undermining, and goals of treatment.[28]

Other factors to consider are dressings that form an effective bacterial barrier, conform to the ulcer shape, cause minimal pain during dressing changes, are nontoxic to the ulcer bed, maintain optimal ulcer temperature, and lower the wound pH.[152] With any ulcer care product, the skin surrounding the ulcer should be monitored for erythema, swelling, pain, or maceration. Erythema, swelling, and pain might be caused by unrelieved pressure or adverse reactions to ulcer care treatments. Skin maceration caused by prolonged contact of ulcer fluid with the skin is a sign

TABLE 25.3 **Examples of Ulcer Dressings**

Type of Dressing	Properties	Examples of Use
Hydrocolloid	Occlusive, adhesive dressing used to maintain moist environment, softens superficial slough, protective, longer wear time, use caution on fragile skin Regular thickness hydrocolloid dressing can absorb a moderate amount of drainage whereas the extra thin version can only contain a minimal amount of exudate	Stage 1, stage 2, and shallow stage 3 pressure injuries, clean ulcers with moderate to no drainage depending on version used and minimal to no depth or tunneling
Transparent	Thin, transparent, adhesive, semipermeable dressing used to maintain moist environment, protective, long wear time, use with caution on fragile skin The absorbent version can be used for ulcers with moderate drainage such as donor sites	Stage 2 and shallow stage 3 pressure ulcers, clean ulcers with minimal to no drainage and no depth or tunneling
Hydrogel	Nonadhesive and nonocclusive Available in sheet form and viscous gel Provides moisture	Used to provide moisture to dry ulcers May be used with other products to pack ulcers
Alginate/hydrofiber	Highly absorbent, can decrease the number of dressing changes with heavily draining ulcers	Use with heavily draining ulcers Also available impregnated with silver to assist with infected ulcers or prevention of infection
Foam	Nonadhesive Provides absorption for ulcer drainage Can reduce friction and shear with some products in this group	Used for moderate to heavily draining ulcers. Can be used as primary or secondary dressing Some layered products can assist with reducing friction and shear and can be used preventatively over areas at high risk for pressure injury
Silver • Impregnated dressings • Silver sulfadiazine	Antimicrobial and is incorporated in many dressing types	More silver does not equate to better antibacterial activity
Cadexomer iodine	Antibacterial and removes bacteria from ulcer surface Absorbs exudate and loose slough	Showing promise in reduction and control of biofilm[167]
Gauze	Some absorption, used to fill dead space, inexpensive	Can be used independently in ulcers with moderate exudate or in conjunction with other products for drier ulcers Use caution not to allow gauze to dry in ulcer bed because this can be painful to remove and is nonselective, meaning it disrupts both healthy granulation tissue and detrimental slough
Collagen matrix	Collagen protein that gives skin its tensile strength	Helpful in all phases of ulcer healing, partial and full thickness, minimal to heavy exudate chronic stalled in inflammatory phase, granulating or necrotic
Composite	Combination dressings	Multiple uses
Cyanoacrylate-based monomer	A nonstinging cyanoacrylate-based monomer that adheres to itself and the skin. It is topical liquid glue that seals off what it is applied to. It forms a flexible and strong protective layer over intact and damaged skin. It resists denudement from caustic fluids, decreases friction, decreases patient discomfort, and may prevent mechanical stripping of skin caused by adhesives	Stage 1 and stage 2 pressure injuries

Data from McNichol LL, Ratliff C, Yates S. Wound, Ostomy, and Continence Nurses Society, Core Curriculum Wound Management. 2nd ed. Wolters Kluwer; 2022; Bryant RA, Nix D. Acute & Chronic Wounds, Current Management Concepts. 6th ed. Elsevier; 2023.

that the topical ulcer care treatment selected is inappropriate for the patient.

Ulcer dressings are used to augment specific ulcer characteristics, making them as ideal as possible. The ideal dressing can only be selected after properly assessing the ulcer and knowing ulcer dressing capabilities. A review of the most commonly used ulcer dressings is discussed later in this chapter.

Gauze

Gauze has become the most widely used surgical dressing. It is inexpensive, readily available, highly permeable, highly absorbent, and nonocclusive. Gauze comes in woven and nonwoven forms and can be used as a primary or secondary dressing. Rolled gauze should be applied in a way to prevent a tourniquet effect. Packing strips are used to prevent premature closure or wick away exudate. Gauze is used on both infected and noninfected ulcers and large or irregularly shaped ulcers. So-called wet-to-dry gauze dressings should be avoided because they compromise wound granulation tissue development and dramatically slow chronic wound healing. An ulcer bed needs to maintain proper and balanced moisture to avoid damaging healthy tissue when removing a dressing. Although wet-to-dry dressings are effective at mechanically debriding an ulcer, removing dried-out gauze can reinjure an ulcer, resulting in pain and delayed ulcer healing. Gauze residue can also cause granuloma formation.[152] In addition, mechanical debridement can cause cross-contamination of ulcers by dispersing bacteria into the air upon removal and indiscriminately debride away adjacent healthy tissue. Gauze is permeable to exogenous bacteria and is associated with a higher infection rate than transparent films or hydrocolloids.[101] Gauze can also cause localized tissue cooling through evaporation, resulting in vasoconstriction, hypoxia, and impaired leukocyte and phagocyte activity, all of which can impair ulcer healing.[101] Because gauze dressings generally require changing three times a day, this can affect nursing time and costs. Most of the criticism regarding the use of gauze, however, remains theoretic. There is a lack of RCTs showing a statistically significant difference between dressings when the primary endpoint is time to ulcer healing.[152]

Impregnated Gauze

These gauze dressings are impregnated with petroleum, iodine, bismuth, silver, or zinc, which make them nonadherent and moderately occlusive. Bismuth can be cytotoxic to inflammatory cells and may augment an inflammatory response. Iodine-impregnated gauze is also cytotoxic and only mildly antimicrobial. These are indicated for tunneling ulcers with foul discharge but must be changed frequently and may damage healthier tissue with prolonged use. They add moisture to the ulcer bed and facilitate healing by decreasing trauma and preventing ulcers from drying out. They can be used as a primary or secondary dressing. They are commonly used on skin graft donor sites, over grafted skin, and on burn ulcers because removal is generally pain free. Impregnated gauze does not absorb exudate and therefore should not be used in heavily exudative ulcers.

Semiocclusive Dressings

Semiocclusive dressings decrease moisture loss from the ulcer and prevent local cooling. Semiocclusive dressings do not significantly reduce costs or ulcer healing time compared with gauze dressings,[160] and the savings from less frequent dressing changes may not justify the higher costs of these products.

Transparent Film

Transparent film dressings are thin, flexible, polyurethane sheets with an adhesive backing. They are permeable to water vapor, oxygen, and carbon dioxide but impermeable to bacteria and water. They provide a moist healing environment and promote autolytic debridement. They do not have any absorptive properties and should not be used in heavily exudative ulcers. They should not be used in infected ulcers because they can promote an ideal environment for bacterial growth. They are primarily used to cover closed surgical incision sites, superficial ulcers with minimal exudate, skin graft donor sites, intravenous catheter sites, areas of friction, and ulcers left to heal by secondary intention. There are also variations of transparent dressings that have absorption features for small to moderately draining ulcers.

Foam Dressings

Foam dressings are polyurethane based and are permeable to gases and water vapor. Their hydrophilic properties make them highly absorptive while providing thermal insulation.[152] They are atraumatic to periulcer tissues upon removal. They are indicated for moderate to heavily exudative, granulating, slough-covered, partial- and full-thickness ulcers, donor sites, ostomy sites, minor burns, and diabetic foot ulcers. They are not recommended in dry, eschar-covered ulcers because they can further desiccate an ulcer. They can remain in place for 4 to 7 days, but should be changed once they are saturated with exudate. If changed daily, they can be used on infected ulcers.[148]

Hydrogels

Hydrogels are complex hydrophilic products that are available as amorphous gels or flexible sheets. They can absorb a minimal amount of fluid, but they primarily add moisture to a dry ulcer, thereby facilitating autolytic debridement, maintaining a moist ulcer environment, and promoting granulation. They are permeable to gas and water but provide a less-effective bacterial barrier compared with occlusive dressings. Because of their high water concentration, they cannot absorb heavy drainage. They can be used on pressure injuries, partial- and full-thickness ulcers, and vascular ulcers. Maceration can be an issue, so periulcer skin may need to be protected with a barrier cream. Hydrogels can be used with topical medications or antibacterial agents. They need to be covered with a secondary dressing and can remain in place for up to 3 days. Hydrogel sheets should not be used on infected ulcers.

Hydrocolloid Dressings

Hydrocolloid dressings comprise a self-adhesive layer of hydrophilic colloid particles, such as carboxymethylcellulose (CMC), pectin, or gelatin. They absorb exudate and swell into a gellike substance over the ulcer, providing a moist healing environment and thermal insulation. The outer polyurethane layer seals and protects the ulcer from bacteria, debris, and shearing. They are available in a variety of shapes and sizes, and they come in paste, powder, or granular form. They prevent contamination, promote autolytic debridement, and do not require a secondary dressing. They can remain in place for up to 7 days or until drainage is noted from beneath the dressing. They are indicated for partial- and full-thickness ulcers with low to moderate exudate, granular or necrotic ulcers, minor burns, and pressure injuries. They cannot be used on infected ulcers. The adhesive can damage fragile periulcer skin.

Alginates

Alginates contain alginic acid from seaweed and are covered in calcium/sodium salts. They are highly absorbent, nonadherent, biodegradable, and some products contain controlled-release ionic silver. They are indicated for highly exudative ulcers. Alginate dressings are available in sheet form, which is beneficial for superficial ulcers, but they are also available in rope form, which is useful for packing deeper ulcers and cavities. To pack an ulcer, fill the dead space, but do not overpack (with gauze, hydrofiber, or alginates). When placed on an ulcer, the sodium and calcium ions interact with serum to form a hydrophilic gel. Alginates provide a moist ulcer environment and can prevent microbial contamination. They are especially effective for managing heavily draining wounds. They are capable of absorbing 20 times their weight, although this varies between products.[152] They are indicated for pressure injuries, vascular ulcers, surgical incisions, wound dehiscence, tunneling wounds, sinus tracts, skin graft donor sites, exposed tendons, and infected wounds. They also have hemostatic properties, which make them suitable for bleeding wounds. They are contraindicated for dry wounds because they are unable to provide any hydration. In clean wounds, they may be kept in place for up to 7 days or until the gel loses its viscosity. For infected wounds, alginate dressings should be changed once daily.

Hydrofiber Dressings

Hydrofiber dressings are made from sodium CMC and are structurally similar to alginates. They form a gel once they interact with serum or exudate.[68] They are easy to remove and are amenable to heavily exudating or infected ulcers. They reduce matrix metalloproteinases (MMPs) and ulcer bioburden because of their highly absorptive quality. Some products also come silver impregnated. They may be left in place for 3 to 7 days. These products are easy to apply, less painful to the ulcer, and reduce nursing costs associated with frequent dressing changes.

Hydroconductive Dressings

Hydroconductive dressings provide capillary action that moves exudate and debris away from an ulcer into the core of the dressing, where it is dispersed into a second layer. These dressings can hold up to 30 to 50 times their own weight.[152] They help loosen adherent slough, allowing for easy removal when the dressing is changed. They can be trimmed to fit a variety of ulcer shapes and sizes. They decrease ulcer exudate, decrease bacterial biofilm, decrease MMPs, and facilitate ulcer bed preparation. Given their ability to absorb high-protein fluid from the ulcer bed, they help reduce bioburden. These dressings do not shed fibers or break apart and can remain in place for up to 7 days.

Silver Dressings

Silver has long been used for its antimicrobial properties. It has broad-spectrum antimicrobial properties and is available in almost every dressing type. Silver cations are released upon contact with fluid. Once released, silver disrupts cell walls, inactivates bacterial enzymes, and interferes with bacterial DNA synthesis. Ionic silver is released at different rates depending on the dressing being used. They effectively reduce bacteria, fungi, viruses, yeast, methicillin-resistant *Staphylococcus aureus* (MRSA), and vancomycin-resistant enterococci (VRE) when used in appropriate concentrations.[165] While decreasing the bacterial count in colonized or critically colonized ulcers, they have no effect on bacteria that have penetrated an ulcer bed. The minimum concentration of silver for most clinically relevant bacteria is between 5 and 50 ppm.[152]

Honey-Based Dressings

Manuka honey has been used to treat infected ulcers for centuries. Several published reports have described the effectiveness of honey products in ulcer healing. In laboratory studies, Manuka honey has been shown to provide antibacterial action against a broad array of bacteria and fungi, including *S. aureus*, *Pseudomonas aeruginosa,* MRSA, and VRE, and can inhibit biofilms of various species.[18] Honey reduces malodor, has autolytic debridement properties, and has antiinflammatory and immune-modifying effects on ulcers.[152] In a retrospective study, Gupta et al. compared honey dressings to silver sulfadiazine dressings for ulcer healing in burn patients. The average time to heal was 18.16 days in the honey group vs. 32.68 days in the silver sulfadiazine group. All ulcers treated with honey became sterile within 21 days, while those treated with silver sulfadiazine became sterile in 36.5 days.[77] They concluded that honey dressings make the ulcers sterile in less time, enhance healing, and have better outcomes regarding hypertrophic scars and postburn contractures.[77]

Iodine-Based Dressings

Iodine is a natural, nonmetallic element essential for human metabolism. Because of their antimicrobial properties, iodine-based products have been used to prevent surgical site infections for several decades. Iodophors were developed in the 1950s to overcome pain and skin irritation associated with elemental iodine. Binding iodine with another molecule makes it less toxic when released slowly over a sustained period of time. The two most common iodophors used today are povidone-iodine and cadexomer iodine. Using povidone-iodine in ulcer healing has been controversial because of alleged issues with toxicity, systemic absorption, and delayed healing. Numerous reviews have analyzed the conflicting evidence. Although animal model studies support the argument for iodine's cytotoxicity, human studies suggest that povidone-iodine preparations enhance ulcer healing by reducing bioburden.[101] Its mode of action is not fully understood but may be caused by its ability to rapidly penetrate a microorganism's cell wall. Povidone-iodine and cadexomer iodine have been shown in numerous in vitro studies to be effective against common ulcer bacteria, including MRSA infections. Slow-release iodine dressings are indicated where infection is suspected, including pressure ulcers, venous leg ulcers, diabetic foot ulcers, minor burns, and superficial skin-loss injuries. To avoid toxicity and thyroid-related complications, iodine products should be used with caution in children and in patients with large burns or large open ulcers. Generally, when the slow-release iodine has lost its color, the antiseptic effect has been lost, and the dressing should be changed.

Charcoal Dressings

Leg ulcers and fungating lesions are more commonly associated with malodor caused by aerobic and anaerobic bacteria.[152] The most effective way of eliminating ulcer odor is removing the causative organism. Systemic antibiotics may be effective, but achieving an effective antibiotic concentration at an ulcer site may be difficult. Activated charcoal dressings deodorize ulcers by absorbing gases released by ulcer bacteria.

Cellular, Acellular, and Matrixlike Products

Of all the advanced ulcer care products, these are the most carefully studied because of the necessity to obtain US Food and Drug Administration approval through randomized trials for patient application and reimbursement.[152] These products are bioengineered from human or animal tissue cells that stimulate fibroblasts and keratinocytes to augment ulcer healing. Acellular products provide a scaffolding to facilitate cell migration to facilitate ulcer healing. Initially developed for burn care, they became more widely used to treat chronic venous and diabetic ulcers. These products closely resemble native epidermis and provide a durable dermal layer for granulation tissue formation. Cellular, acellular, and matrixlike products[170] require excellent wound bed preparation (debridement, moisture control) to maximize outcomes,[64,162] and they cannot be used with active osteomyelitis, cellulitis, or ulcer site infection. They also cannot be placed over necrotic tissue, draining ulcers, or exposed bone.

Modalities

Hyperbaric Oxygen Therapy

Although enhanced ulcer healing is a postulated benefit of adjunctive therapy with hyperbaric oxygen, the paucity of RCTs makes it difficult to assess the efficacy of hyperbaric oxygen. A trial of adjunctive therapy with hyperbaric oxygen at 2.0 to 2.5 atmospheres for 90 to 120 minutes can be considered for patients with ischemic ulcers if the periulcer transcutaneous oxygen tension is shown to increase during treatment.[158] Predictive models exist that may be useful in selecting the patients most likely to benefit from hyperbaric oxygen therapy.[59] Current covered wound indications include[154]:

- Compromised surgical grafts and flaps
- Crush injuries/skeletal muscle compartment syndrome/acute arterial insufficiency
- Necrotizing soft tissue infections
- Specific acute thermal burns and frostbite
- Delayed radiation injuries for soft tissue or bony necrosis/osteoradionecrosis
- Chronic refractory osteomyelitis
- Enhancement of healing in a problematic wound (diabetic foot ulcers Wagner grade 3, 4, or 5)

Noncontact Low-Frequency Ultrasound

The goals of noncontact low-frequency ultrasound include reducing the bacterial bioburden and enhancing granulation tissue formation. A retrospective observational study of 163 people with vascular ulcers showed a significantly greater proportion of ulcers treated adjunctly with noncontact, low-frequency ultrasound healed compared with the standard of care alone. Among the ulcer subgroups studied, venous ulcers seem to benefit the most from adjuvant noncontact low-frequency ultrasound, although improvements were also observed for ischemic and neuropathic ulcers.[95]

Electrical Stimulation

Electrical stimulation has been demonstrated to enhance ulcer healing through effects on blood flow, edema reduction, antibacterial properties, cellular migration, fibroblast activity, and ulcer tensile strength.[100] The mechanisms through which electrical stimulation exerts a positive effect on pressure ulcer healing are reasonably well established. Further studies to investigate the use of electrical stimulation in different clinical settings are needed.[96]

Negative Pressure Ulcer Therapy

Negative pressure wound therapy (NPWT) applies subatmospheric pressure to an ulcer bed via a suction device attached to a sponge.[125] Closed suction removes excess fluid from the ulcer, allowing for enhanced circulation, removal of cellular waste, and reduced risk of bacterial contamination. It helps manage extensive ulcers that cannot be addressed with primary closure or are too large to be treated with conservative ulcer strategies. NPWT augments granulation and epithelialization. An adhesive drape over the sponge provides a semiocclusive environment that encourages moist ulcer healing and facilitates gas exchange. Black polyurethane foam has large pores and is more effective for stimulating granulation tissue and ulcer contraction. White polyvinyl alcohol foam is denser with a smaller pore size and is recommended for deep ulcers, undermined flaps, sinus tracts, or overexposed bone. It is less likely to adhere to the ulcer bed. White foam does not stimulate granulation. Dressing changes can vary between 48 and 72 hours, but untreated infected ulcers should be changed every 12 hours.

Summary

Chronic ulcers are a challenge regardless of the underlying etiology. Identification of patients at high risk for developing ulcers can direct prevention strategies to those most likely to benefit. Once an ulcer occurs, treating the underlying medical condition, consistent offloading, and creating an optimal ulcer healing environment are all key to ulcer healing and avoiding recurrence.

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